

Supplementary Information

Tunable Nanostructuring for van der Waals Materials

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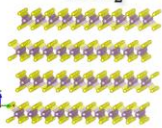
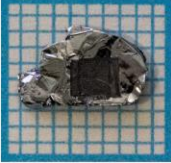
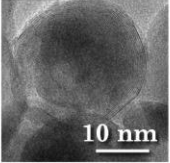
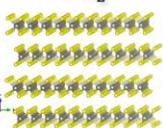
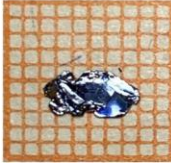
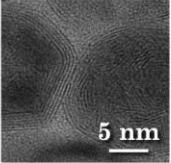
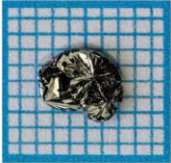
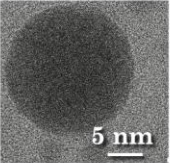
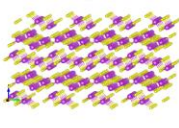
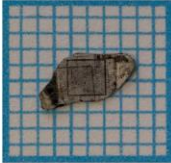
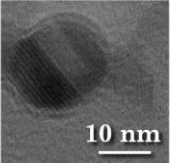
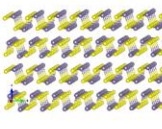
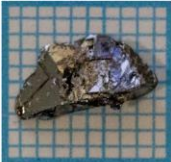
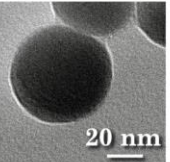
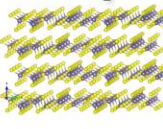
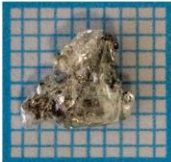
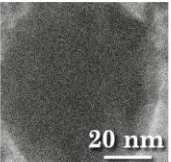
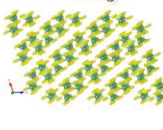
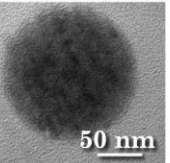
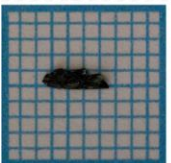
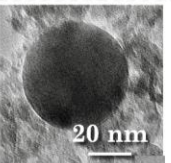
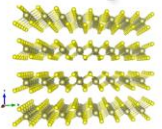
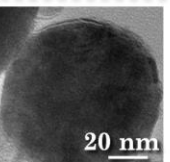
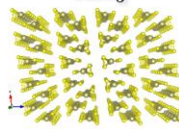
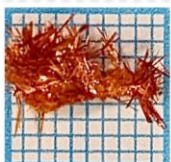
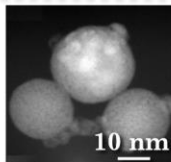
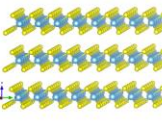
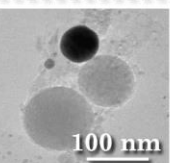
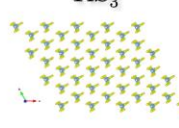
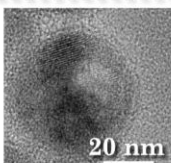
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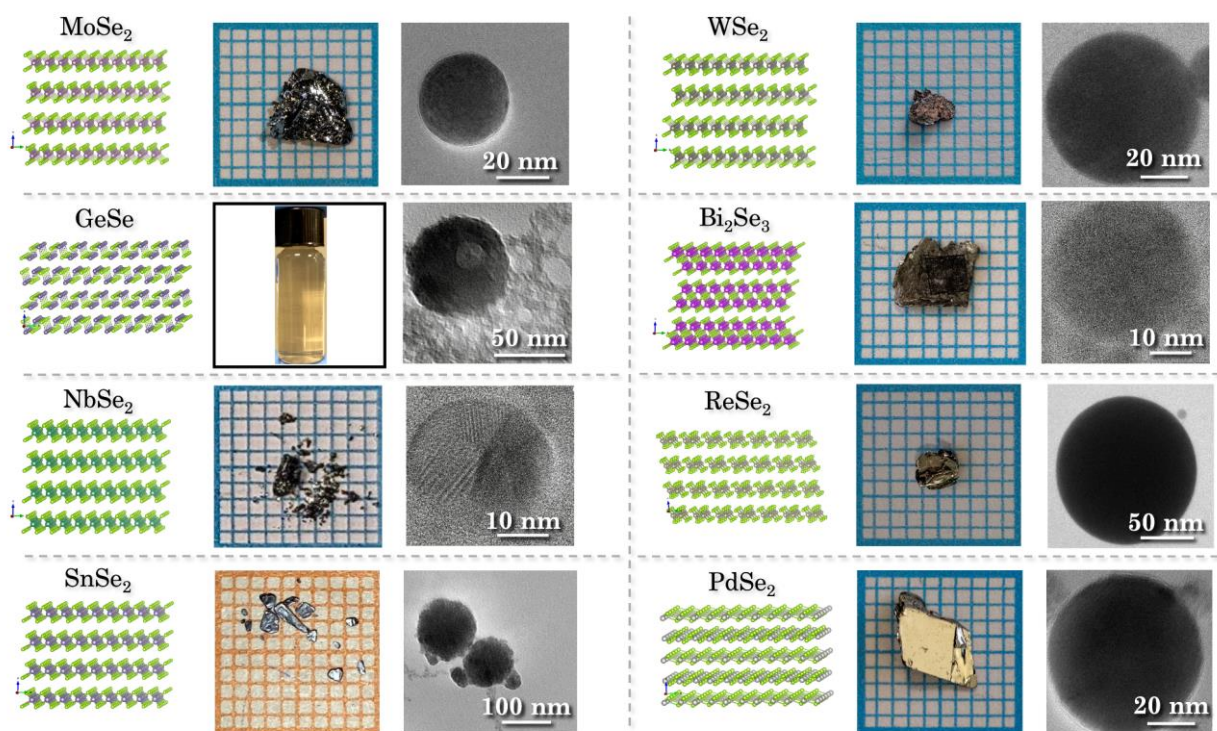
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Sulfides

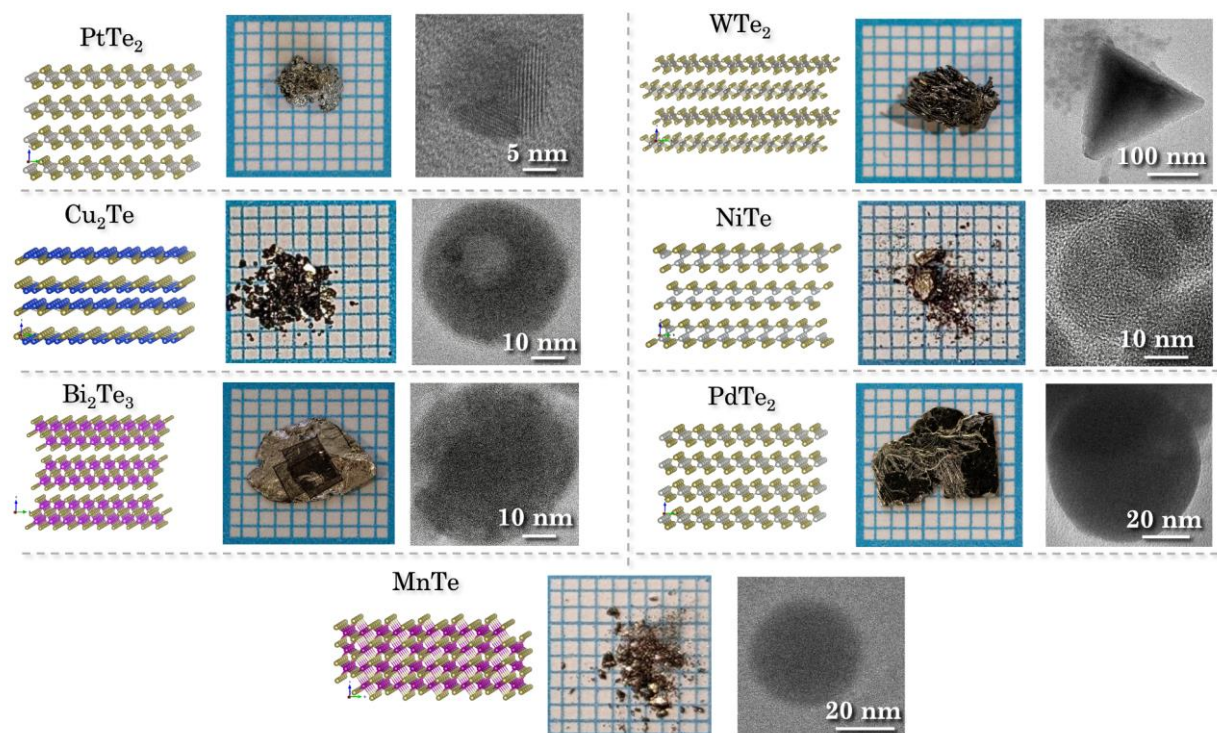
MoS_2 			WS_2 		
ReS_2 			Bi_2S_3 		
GeS 			GeS_2 		
NbS_3 			TaS_2 		
HfS_2 			HfS_3 		
TiS_2 			TiS_3 		

Selenides



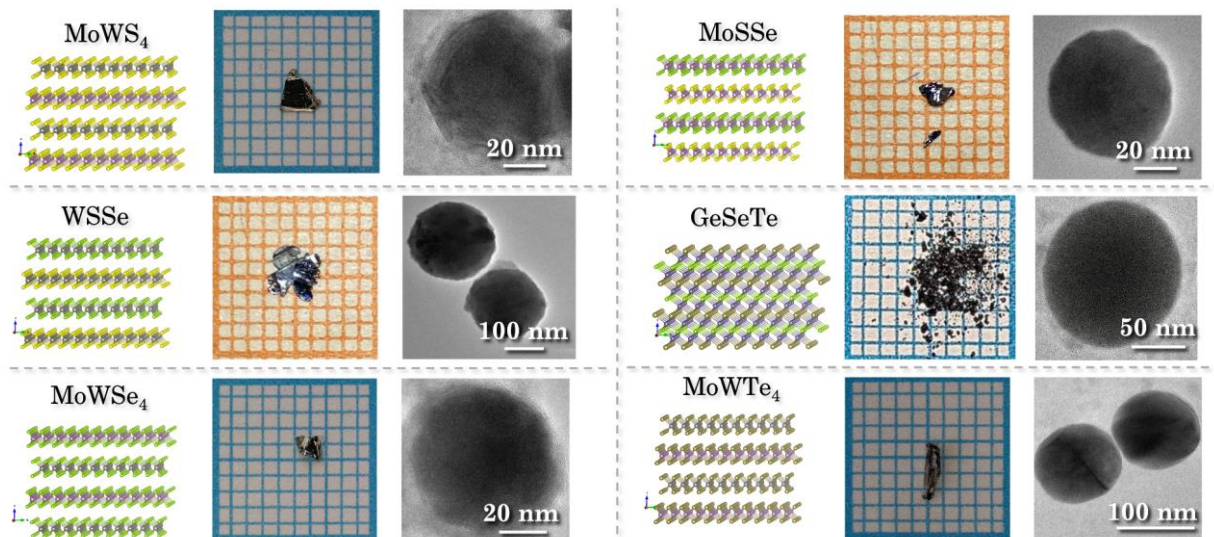
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Tellurides



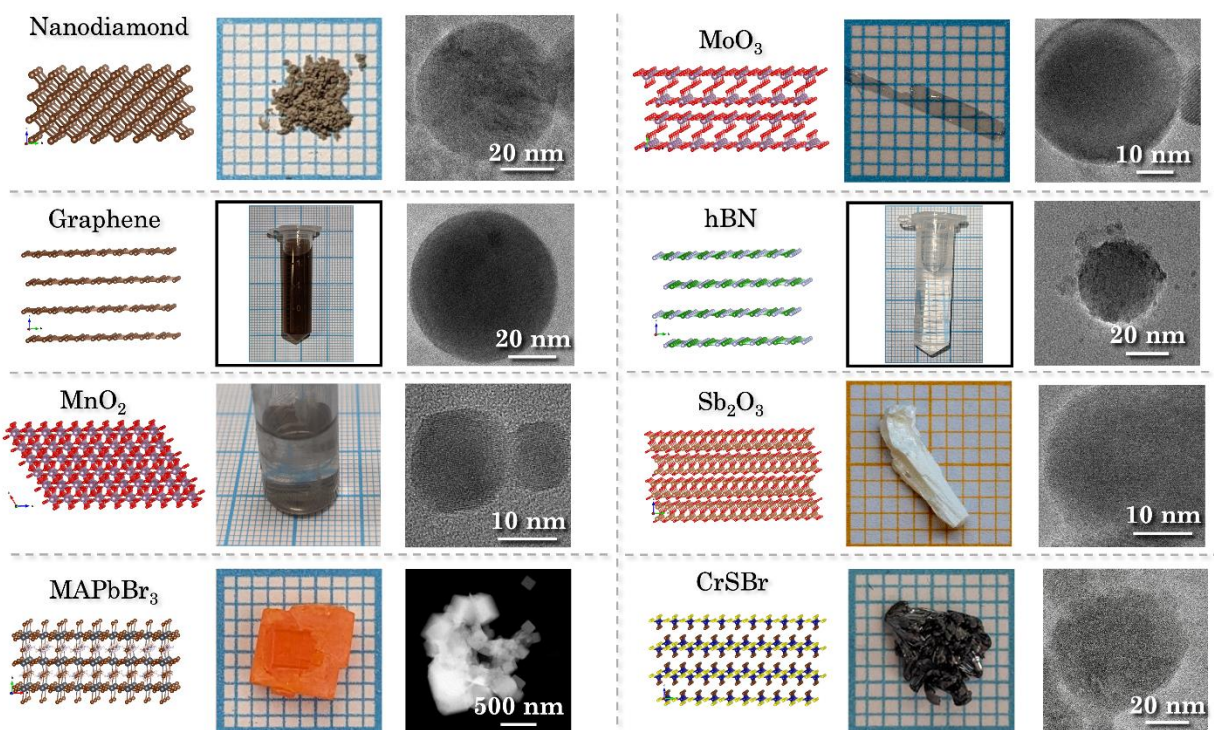
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Janus



26

Others



27

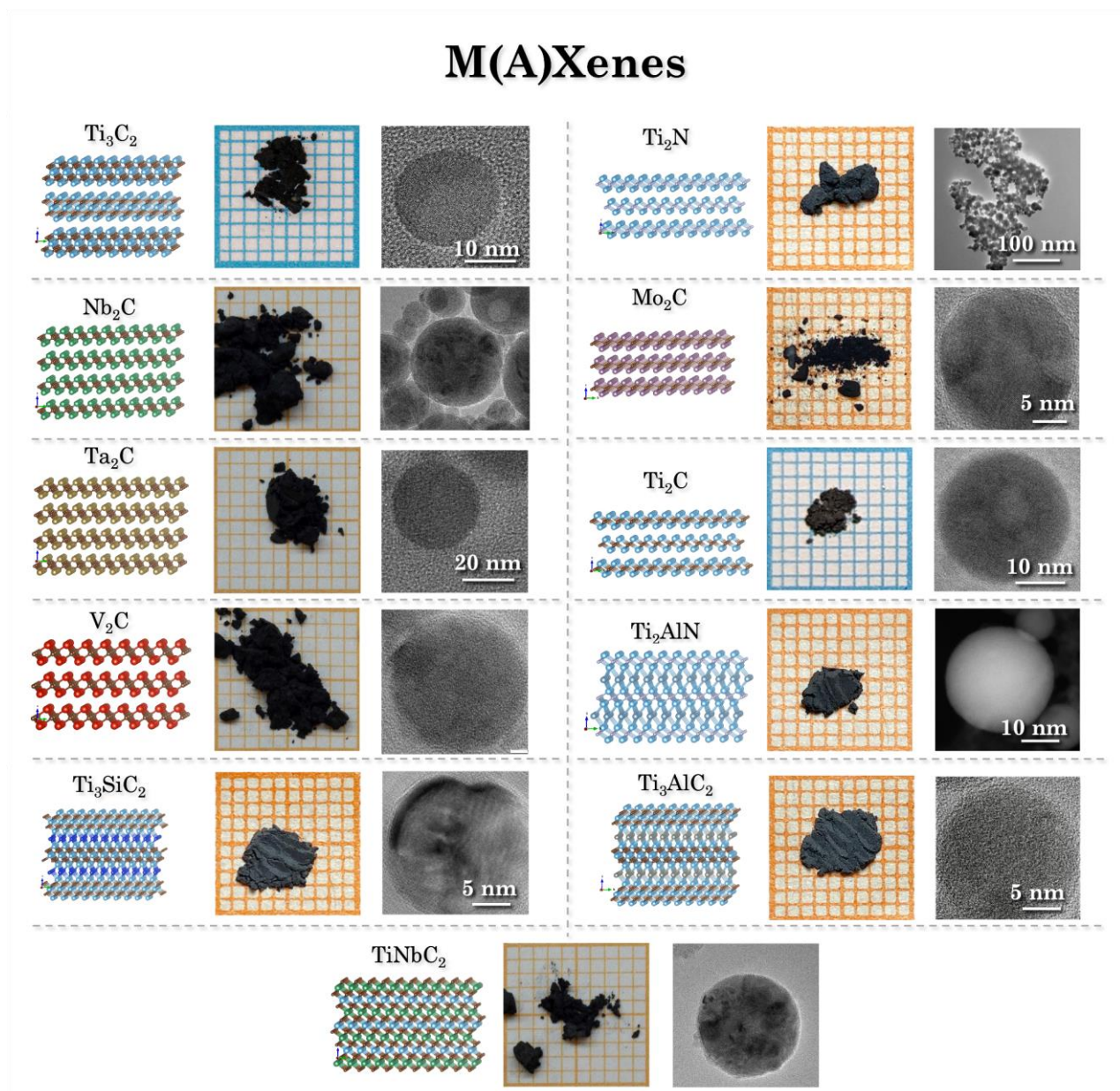


Figure S1. Three-dimensional representation of the crystal structure of vdW materials, pictures of initial crystals/powders/solutions, and TEM image of laser-synthesized vdW NPs divided into different material families (TMDCs: Sulfides, Selenides, Tellurides, Janus, M(A)Xenes, Others).

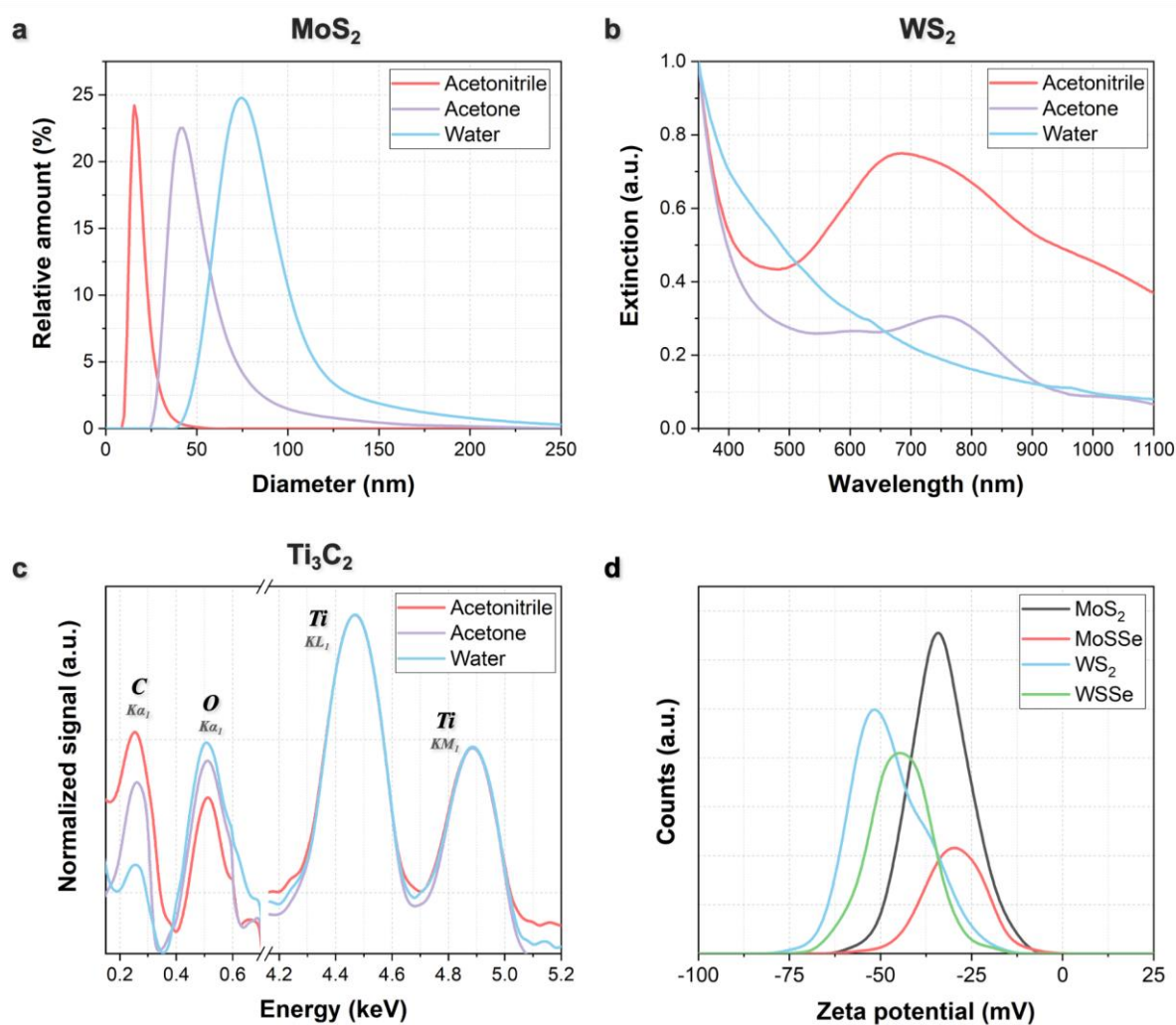


Figure S2. Dependence of (a) size, (b) optical, and (c) compositional properties of MoS₂, WS₂, and Ti₃C₂ NPs on liquid type, respectively. (d) Zeta-potential distribution of various laser-synthesized vdW NPs, confirming high colloidal stability.

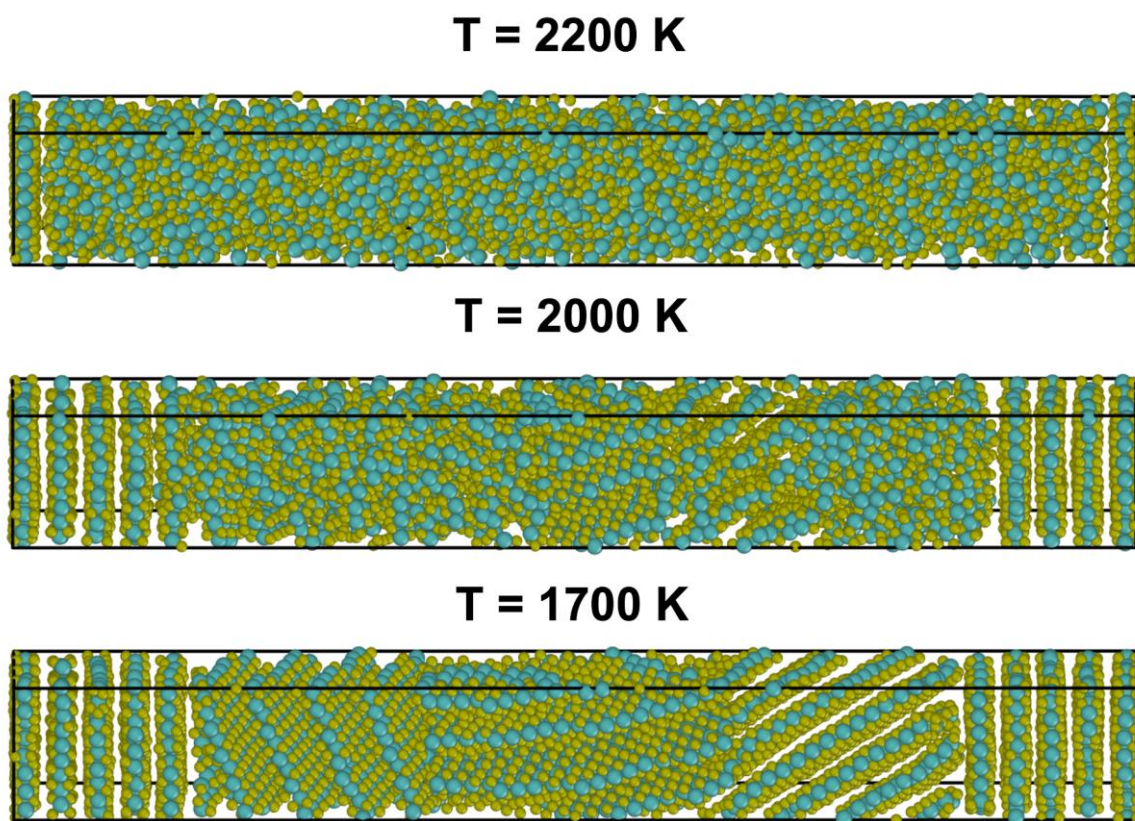


Figure S3. The atomic structure of MoS₂ polycrystalline NP for various temperatures generated in MD.

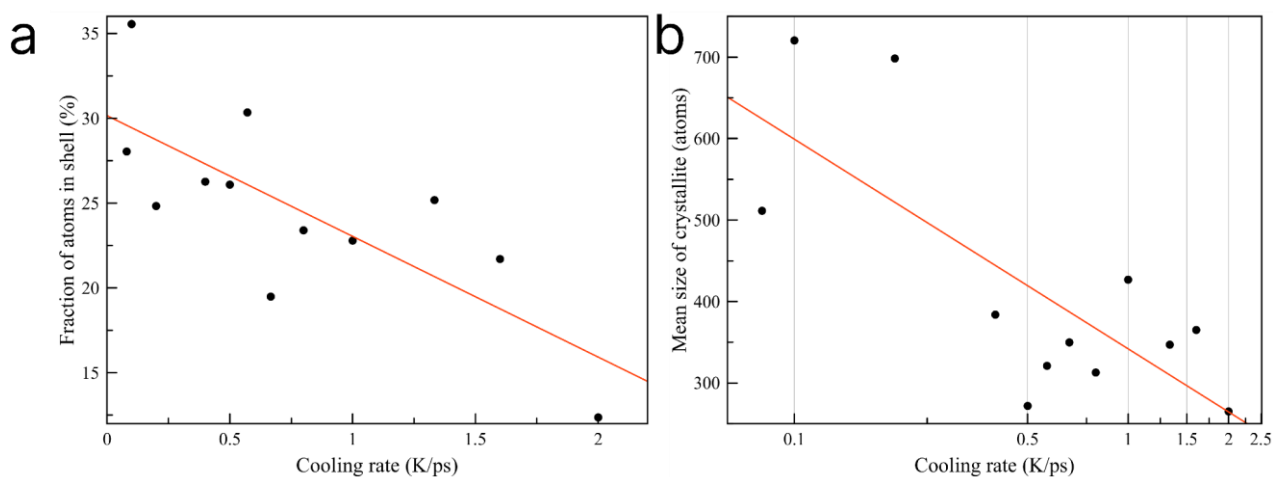


Figure S4. MD calculated dependence of the fraction of all atoms in the NP shell on (a) the average size of the inner crystallite and (b) the cooling rate.

Table S1. Comparison of interplanar distance (d_{exp}) determined from SAED patterns of vdW NPs and spacing between crystallographic planes (d_{hkl}) with corresponding Miller indices (hkl).

Material	Ring	$d_{\text{exp}}(\text{\AA})$	hkl	$d_{\text{hkl}}(\text{\AA})$
MoSe ₂	1	6.76	(0 0 2)	6.77
	2	3.36	(0 0 4)	3.39
	3	2.88	(0 1 0)	2.88
	4	2.46	(1 0 3)	2.43
	5	1.68	(1 1 0)	1.66
	6	1.17	(1 1 8)	1.19
HfS ₃	1	2.87	(0 1 2)	2.82
	2	2.52	(2 0 0)	2.54
	3	1.77	(0 2 0)	1.8
	4	1.49	(2 2 0)	1.47
	5	1.43	(2 2 1)	1.43
PtTe ₂	1	2.91	(0 1 1)	2.95
	2	2.19	(1 0 2)	2.13
	3	2.04	(1 1 0)	2.05
	4	1.72	(2 0 1)	1.68
	5	1.5	(0 2 2)	1.48
WTe ₂	1	2.2	(1 0 5)	2.14
	2	1.9	(1 0 6)	1.93
	3	1.25	(1 0 11)	1.24
	4	1.1	(2 1 5)	1.09
Ti ₃ C ₂	1	2.48	(1 0 2)	2.51
	2	2.15	(1 0 4)	2.17
	3	1.52	(1 1 0)	1.53
	4	1.3	(1 1 6)	1.31
	5	1.24	(2 0 4)	1.26
MAPbBr ₃	1	2.9	(2 0 0)	2.96
	2	5.85	(1 0 0)	5.92
	3	4.1	(1 1 0)	4.19
	4	2.58	(1 2 0)	2.65
	5	2.9	(0 2 0)	2.96
	6	5.8	(0 1 0)	5.92

Supplementary Note 1. Mechanical Modeling of Shell Shape Formation.

We attribute the observed differences in the shell shape of the nanoparticles to the mechanical properties of the monolayer materials—specifically, their bending modulus and resistance to out-of-plane deformation. We hypothesize that materials with a higher bending modulus tend to preserve a spherical shell shape, whereas more easily bendable materials are prone to forming polygonal geometries.

Among the studied systems, MoS₂ and WS₂ nanoparticles exhibit the most pronounced contrast in morphology: MoS₂ shells appear nearly spherical, whereas WS₂ shells adopt polygonal

geometries. Figure S5 shows the dependence between potential energy density and bending curvature. The calculated bending modulus for MoS₂ is approximately $E_b \approx 9.6$ eV, while WS₂ is significantly more flexible, with $E_b \approx 7.3$ eV (see Table S2 for literature values).

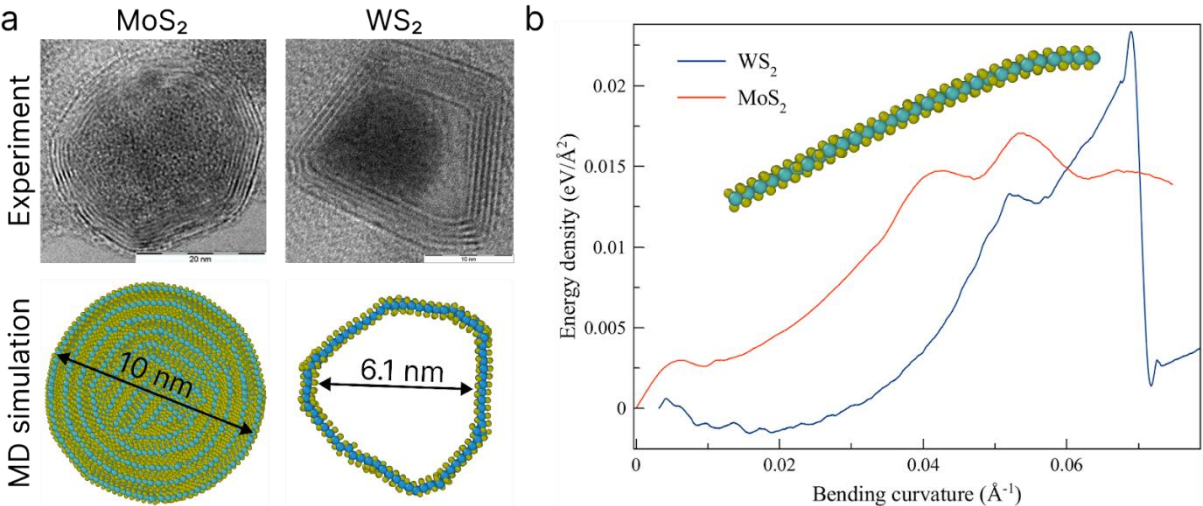


Figure S5. Comparison of elastic properties between MoS₂ and WS₂. (a) Top: TEM images of spherical MoS₂ and polygonal WS₂ nanoparticles. Bottom: relaxed multilayer structures with three-shell and three-core crystalline models, and the cylindrical deformation model. (b) Bending modulus and critical angle analysis for single-layer MoS₂ and WS₂.

Table S2. Bending moduli and critical angles of MoS₂ and WS₂ from this work and literature sources.

vdW material	Bending modulus, eV	Critical angle, °
MoS ₂	9.2* 9.6 ^[1] 9.61 ^[2]	68* 85 ^[4]
WS ₂	7.3* 5.32 ^[3]	57* 58 ^[5]

*this work

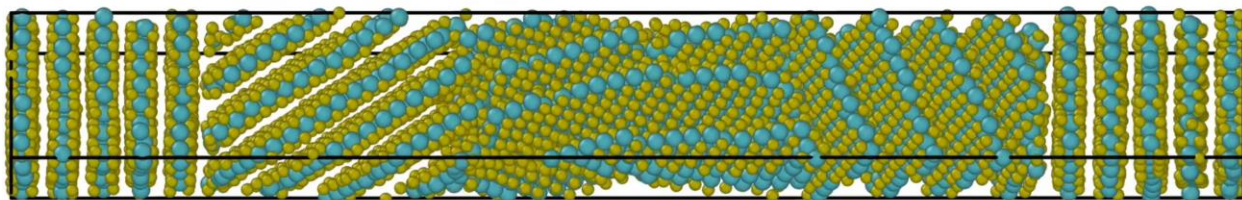
References

- [1] J.-W. Jiang, Z. Qi, H. S. Park, and T. Rabczuk, "Elastic bending modulus of single-layer molybdenum disulfide (MoS₂): finite thickness effect," *Nanotechnology*, vol. 24, no. 43, p. 435705, Nov. 2013, doi: 10.1088/0957-4484/24/43/435705.
- [2] J. H. Kim, J. H. Jeong, N. Kim, R. Joshi, and G.-H. Lee, "Mechanical properties of two-dimensional materials and their applications," *J. Phys. Appl. Phys.*, vol. 52, no. 8, p. 083001, Feb. 2019, doi: 10.1088/1361-6463/aaf465.
- [3] A. Falin et al., "Mechanical Properties of Atomically Thin Tungsten Dichalcogenides: WS₂, WSe₂, and WTe₂," *ACS Nano*, vol. 15, no. 2, pp. 2600–2610, Feb. 2021, doi: 10.1021/acsnano.0c07430.

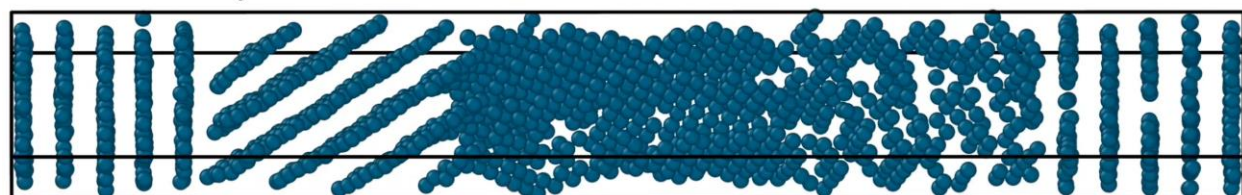
[4] G. Deokar et al., “Synthesis and characterization of MoS₂ nanosheets”, Nanotechnology, vol. 27, p. 075604, Jan 2016.

[5] I. Kaplan-Ashiri et al., “On the mechanical behavior of WS₂ nanotubes under axial tension and compression”, vol. 103, no. 3, pp. 523-528, Jan 2006.

All atoms in computational cell:



Mo atoms in crystal-like environment :



Supplementary Movie 1. A snapshot at 00:36 is shown here. The movie presents the simulation of the self-organization process of the inner polycrystalline structure and shell layers of MoS₂ NPs using Molecular Dynamics.